Machine Learning for the study of glacial-interglacial cycles (TFM)

Experiment A1: SINDy for recovering PACCO equations

Adrián Ordóñez Sanmartín

29/04/2026

What is the plan and what have we already done?

Validation



Real results

Experiment A - SINDy to PACCO with all the variables

A1 - Not noisy, equally spaced.

A2 - Noisy, equally spaced.

A3 - Not noisy, unequally spaced.

A4 - Not noisy, unequally spaced.

Now we try this, to really validate the method.

Experiment B - SINDy to PACCO only with proxy observables (“H”, C, I).

B1 - Not noisy, equally spaced.

B2 - Not noisy, unequally spaced.

In the previous weeks we did this.

We found 100 ky cycles. Poor stability and not that sparse results.

Experiment C - SINDy to proxy data.

We will define this in the future.

A1 - Methodology

We try to recover PACCO equations (for the 5 pronostical variables). In the first approximation, this are its dependencies:

$$
\begin{cases}
dT/dt = F(H, B, H_sed, C, T, I) \\
dC/dt = F(C, T) \\
dH/dy = F(T, \alpha, H^3, I) \\
dB/dt = F(H, B) \\
d(H_sed)/dt = F(T, H^4)
\end{cases}
$$



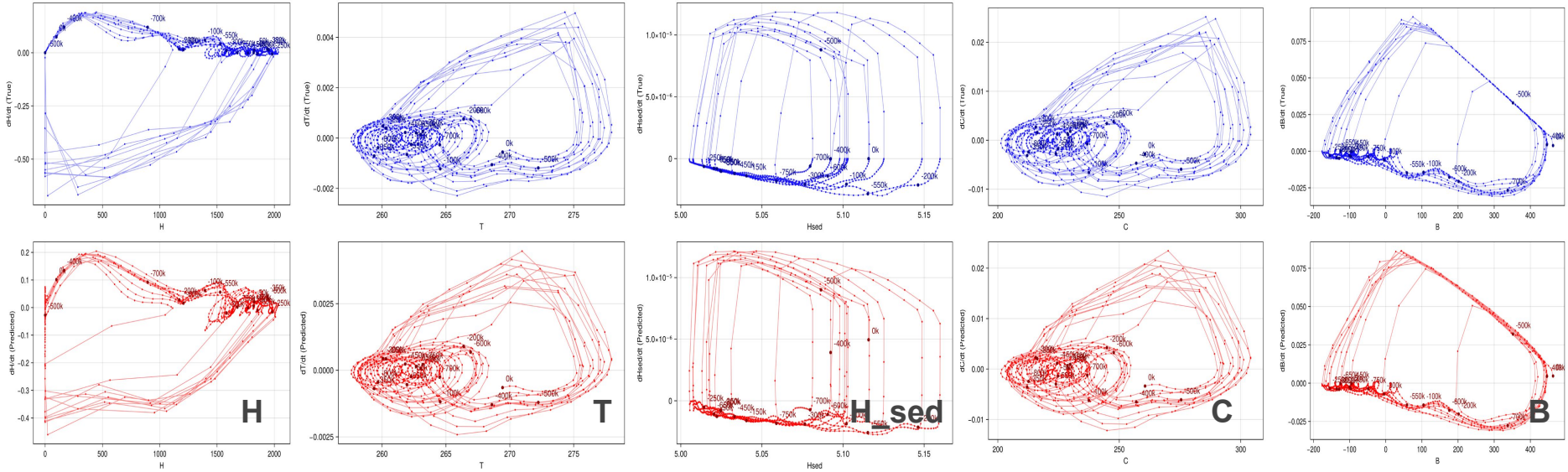
We have not included the albedo variable, since its anomalous behaviour confused SINDy

We also find for hysteresis, using the continuation method.

Result 1 - Best fit. Hysteresis. Not sparse.

Solver: STLSQ ($\lambda = 0.1995$) Equations: 5 Parameters: 52 Independent variable: τ $R^2 = 0.8850$

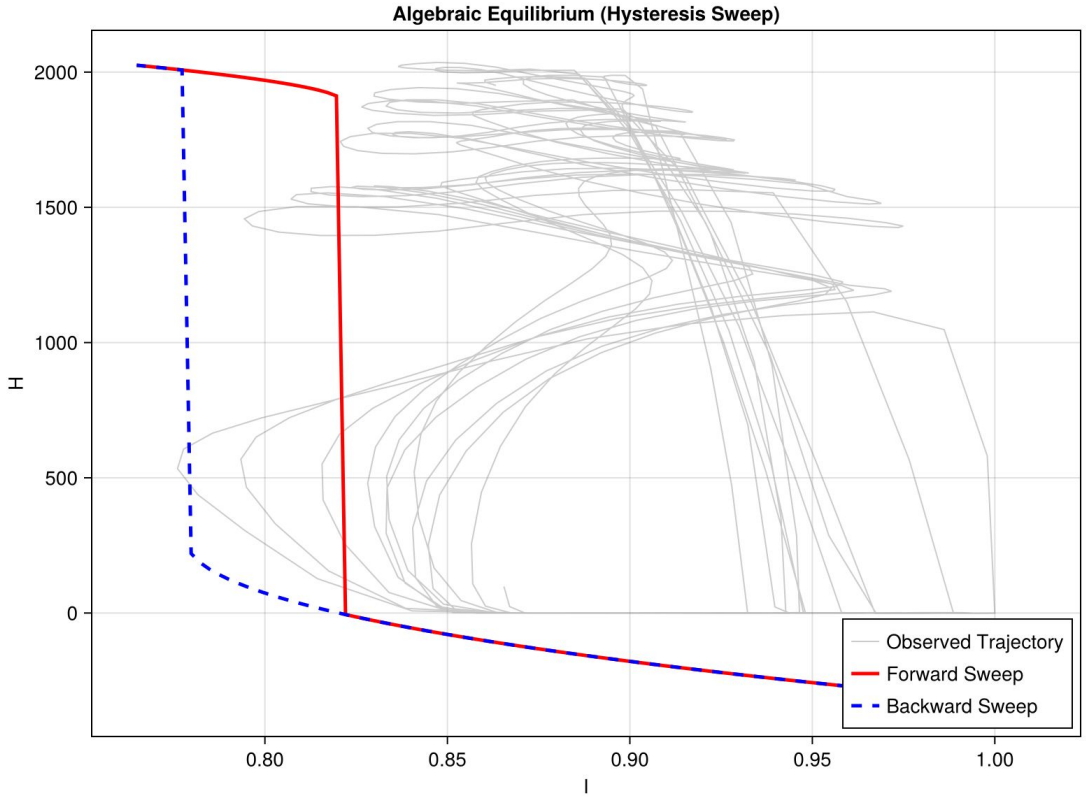
States: H, C, B, H_{sed}, T Input: I $\tau \in [-800\,000,\ 0]$



Phase diagrams for the 5 prognostic variables. In blue, with the numerical derivative. In red, with the discovered derivative

Trajectories are well represented for all variables. No one was stable under integration.

Result 1 - Best fit. Hysteresis. Not sparse.



H*T term made the hysteresis appear

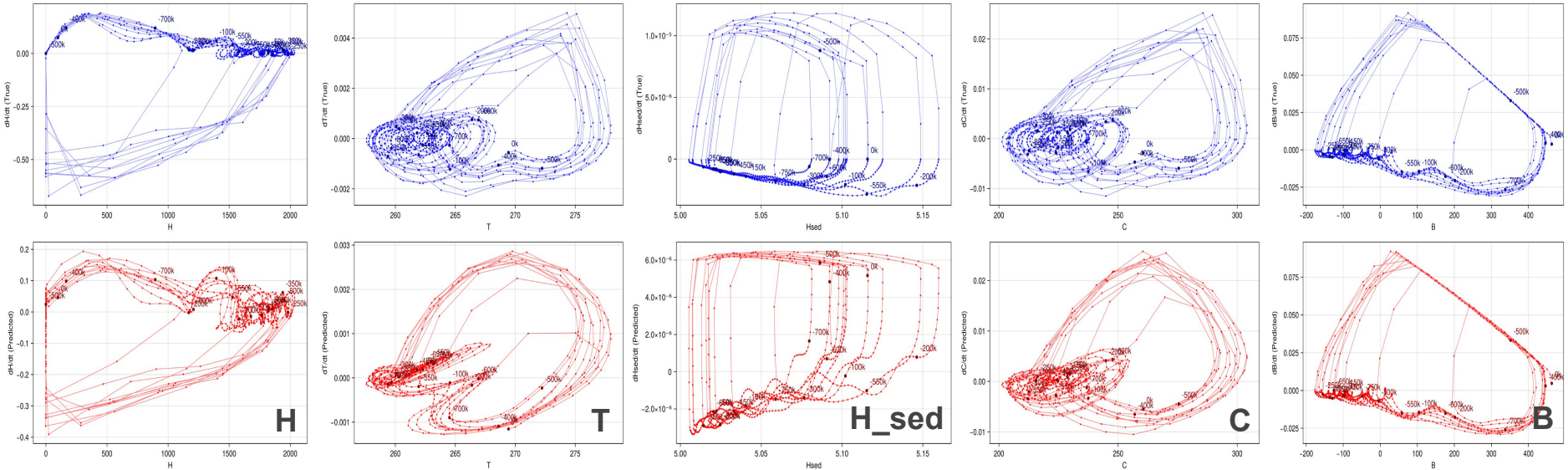
Hysteresis is found for all 5 variables vs I.

For reference:

I = 1 is 565 W/m^2

Result 2 - Sparser, but not hysteretic.

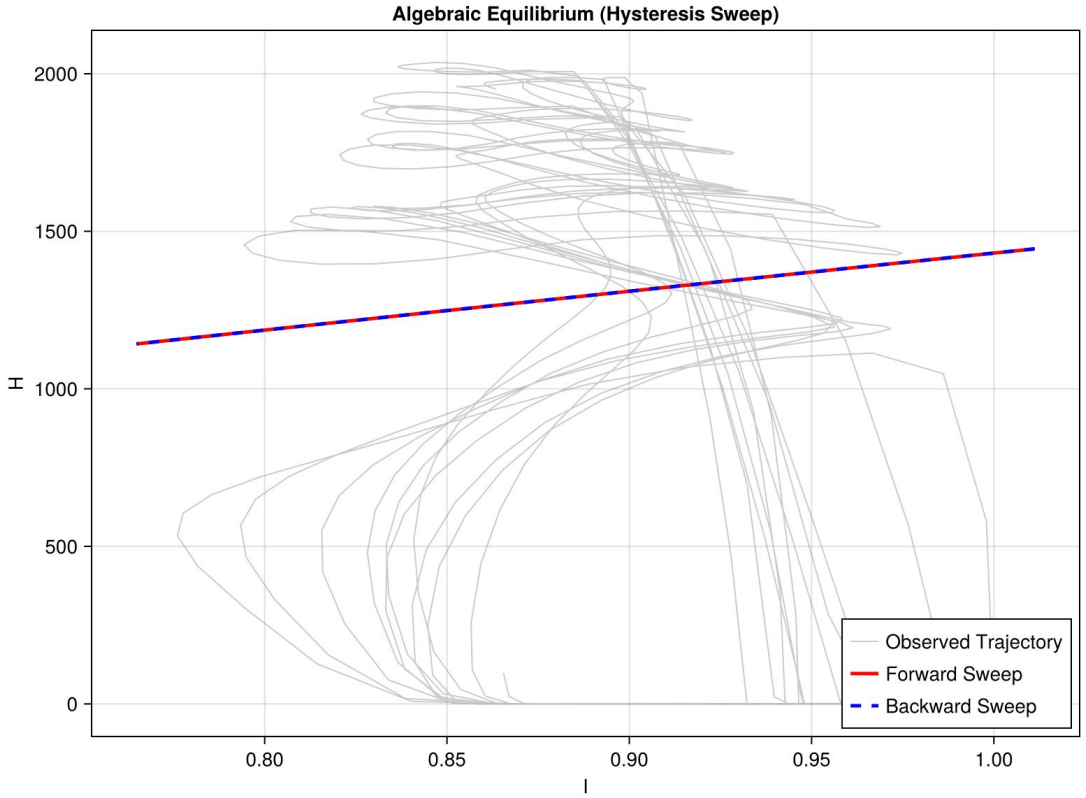
Model 2 (ADMM): *sparse, no hysteresis*, R^2=0.846, 25 parameters, $\tau \in [-800\,\mathrm{kyr}, 0]$.



We have lost some precision. Even the attractor in T. Not even one of the variables was stable over integration

Phase diagrams for the 5 prognostic variables. In blue, with the numerical derivative. In red, with the discovered derivative

Result 2 - Sparser, but not hysteretic.



Hysteresis is lost for all 5 variables vs I.

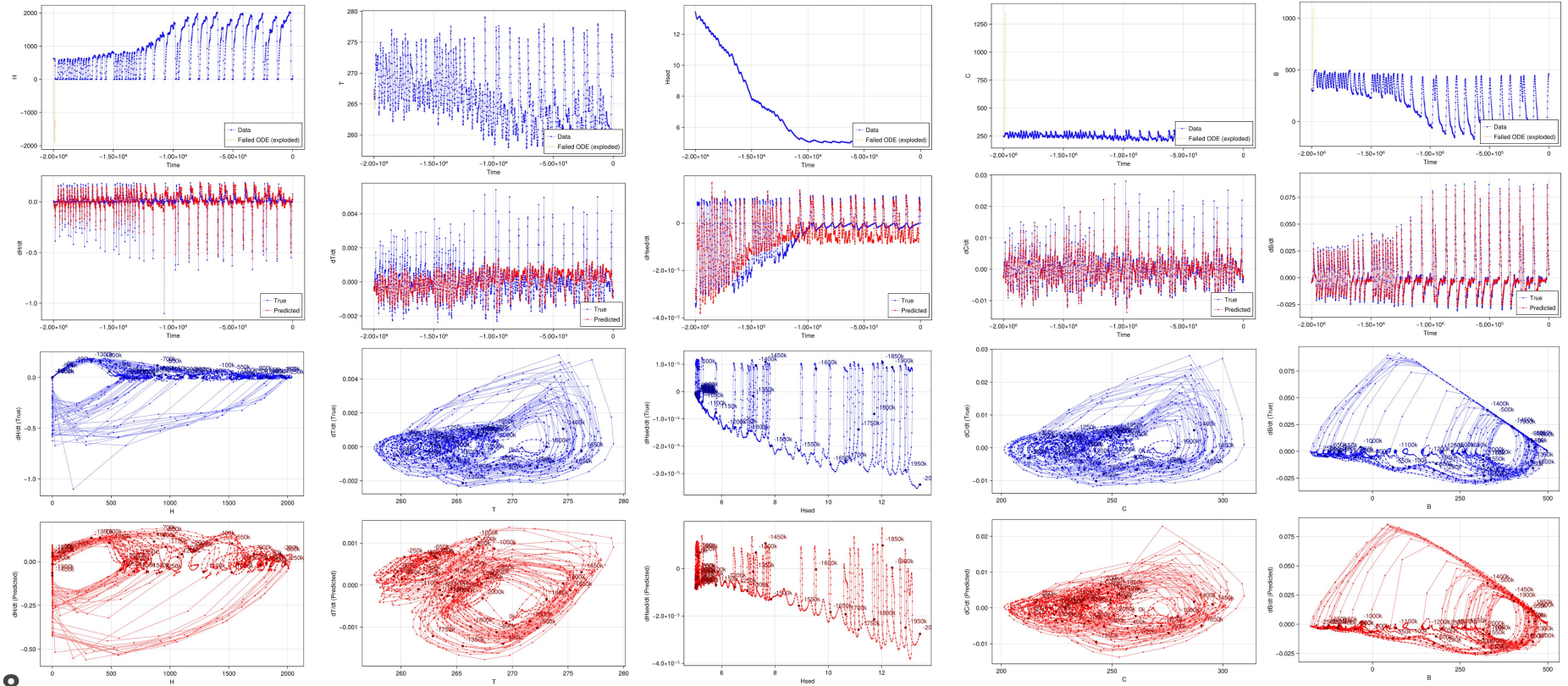
For reference:

I = 1 is 565 W/m^2

Result 3 - Last 2 Myr.

Phase diagrams for the 5 prognostic variables. In blue, with the PACCO data. In red, with the discovered model.

Model 3 (STLSQ, λ=6.31): *all attractors, 2 Myr period,* R^2=0.838, 38 parameters, $\tau \in [-2\,\mathrm{Myr}, 0]$.



Result 1-3 SINDy discovered equations

Table 1: Fitted coefficients for all three SINDy models. Each cell shows the coefficient of the row's basis function in the corresponding state equation. Dashes (—) indicate the term is absent. **Model 1** (STLSQ, λ=0.20): *best fit with hysteresis, not sparse*, R^2=0.885, 52 parameters, $\tau \in [-800\,\mathrm{kyr}, 0]$. **Model 2** (ADMM): *sparse, no hysteresis*, R^2=0.846, 25 parameters, $\tau \in [-800\,\mathrm{kyr}, 0]$. **Model 3** (STLSQ, λ=6.31): *all attractors, 2 Myr period*, R^2=0.838, 38 parameters, $\tau \in [-2\,\mathrm{Myr}, 0]$. States: $H, C, B, H_s \equiv H_{\mathrm{sed}}, T$. Input: I.

	modA**Model 1** — STLSQ (λ=0.20), R^2=0.885					modB**Model 2** — ADMM, R^2=0.846					modC**Model 3** — STLSQ (λ=6.31), R^2=0.838				
Term	$\dot{H}$	$\dot{C}$	$\dot{B}$	$\dot{H}_s$	$\dot{T}$	$\dot{H}$	$\dot{C}$	$\dot{B}$	$\dot{H}_s$	$\dot{T}$	$\dot{H}$	$\dot{C}$	$\dot{B}$	$\dot{H}_s$	$\dot{T}$
1	−809.26	349.47	−117.15	1.58	67.93	—	2.19	—	—	—	−1050.60	303.39	−161.55	15.55	35.74
H	485.74	19.17	53.36	1.86	3.33	73.40	—	−13.86	−0.15	—	201.32	—	10.06	—	—
C	−35.94	8.64	−6.52	0.51	0.38	175.51	—	2.23	0.13	—	76.43	—	23.27	—	—
B	25.40	−5.23	−17.96	—	−1.11	18.76	−7.48	−18.60	—	−1.32	105.97	—	—	—	—
H_s	31.57	15.20	6.45	−3.37	4.57	−42.55	—	—	—	—	153.34	70.47	15.11	−12.77	—
T	878.99	−459.05	151.91	—	−91.16	—	—	19.71	—	—	1093.42	−389.52	183.26	−15.46	−43.01
I	−80.61	90.25	−13.09	—	19.26	−148.12	51.45	—	—	1.05	134.41	−6.71	44.32	−15.05	−27.90
H^2	−411.16	−18.75	−66.63	−2.96	−3.75	−32.96	−20.57	—	—	−1.45	—	—	—	—	—
$T \cdot H$	−351.63	−39.79	−64.71	−1.89	−7.76	—	−3.45	−15.51	—	—	—	—	—	—	—
$C \cdot H$	—	—	—	—	—	—	—	—	—	—	−162.94	−18.41	−35.04	—	—
$B \cdot C$	—	—	—	—	—	—	—	—	—	—	−86.51	—	−19.72	—	—
$T \cdot H_s$	—	—	—	—	—	—	—	—	—	—	−162.57	−74.11	−16.02	12.98	—
$T \cdot I$	—	—	—	—	—	—	—	—	—	—	−260.51	96.20	−67.89	14.98	35.57
H^3	580.28	25.95	94.61	−1.89	5.25	—	—	—	—	—	—	—	—	—	—
H^4	−264.19	−11.96	−43.32	1.45	−2.39	16.50	6.90	0.92	—	—	—	—	—	—	—

Notes. (i) Model 1 includes polynomial nonlinearities up to H^4 plus the interaction $T \cdot H$, yielding good hysteresis reproduction at the cost of 52 parameters. (ii) Model 2 (ADMM) selects a sparser library (25 parameters) but loses the hysteresis cycle. (iii) Model 3 uses bilinear cross-terms ($C \cdot H$, $B \cdot C$, $T \cdot H_s$, $T \cdot I$) instead of high-degree polynomials, covering the full 2 Myr attractor set. $H_s \equiv H_{\mathrm{sed}}$.

Conclusions

Hysteresis found

We have a tool for searching which variables produce the hysteresis

PACCO “kinda of” rediscovered?

It is difficult to say. Of course, we have at least discovered a really good approximation